

M3_Practice

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Key Terms/Concepts:

- **News Vendor Problem** - deciding how much product to order (and sometimes what price to sell it for)
- **Law of Large Numbers** - as sample size increases, sample average will converge towards population mean.
- **Flaw of Averages** - fallacy that the average value can represent individuals within a population. Averages can be misleading because they do not account for variability and distribution of characteristic within population.
- **Sensitivity Analysis** - assessing robustness of model by evaluating impact of changes in key variables on output of model and decisions.

Roll With It Sushi

Renowned for serving the freshest sushi in the city, Roll With It Sushi is hosting a yellowtail festival. Every morning, the sous-chef purchases fish at the market and any unsold fish at the end of the day gets discarded.

The restaurant has contacted you because they are still determining how many people will attend the event and worry about how this will impact their business financially. To ensure they have enough sushi and budget appropriately, they want you to recommend the amount of fish needed. Based on past events, they expect at least 20 people to attend and have the capacity to seat up to 50 guests. They anticipate the most likely attendance to be around 30 people, and without your guidance they feel like this is the best guide in determining the fish needed.

Roll With It Sushi can purchase any quantity of yellowtail from the fresh market at \$9 for 16 ounces (divisible). Each guest at the festival is entitled to five ounces of fish. If the sushi runs out, some customers will not be happy. However, the restaurant has promised to refund their entry fee. Additionally, they have promised to serve their famous miso soup to every attendee. The cost of making a batch for up to 50 guests is \$300.

Given that the festival charges \$20 per entry, how much yellowtail would you recommend the restaurant to purchase to maximize expected profits?

Model Framework

```
library(extraDistr)

# Decision Variable - looking for the optimal value of this variable.
Order_Oz<-160 # Decision Variable

# Inputs - fixed/predetermined values. Gives the model its structure.
Price_Fish_Oz<-9/16 # Input
Price_Miso<-300 # Input
Entry_Fee<-20 # Input
Fish_Entitled_Oz<-5 #Input

# Random Variables - primary source of uncertainty. Often modeled by sampling probability distributions.
set.seed(20)
Attendance<-round(rtriang(1,20,50,30),0) # Random Variable

# Calculated Values - transform inputs and help describe model.
Consumption<-Attendance*Fish_Entitled_Oz # Calculated
Available<-Order_Oz # Calculated

#outcome.
Profit<-min(Consumption,Available)/Fish_Entitled_Oz*Entry_Fee-Order_Oz*Price_Fish_Oz-Price_Miso

#Simulated/calculated attendance.
print(Attendance)
```

```
[1] 44
```

```
#Amount of fish consumed according to simulated attendance.
print(Consumption)
```

```
[1] 220
```

```
#Expected profit.
print(Profit)
```

```
[1] 250
```

Law of Large Numbers

```
#Running the simulation 10,000 times. *Law of Large Numbers
n<-10000
V_Order_Oz<-rep(Order_Oz,n)
V_Price_Fish_Oz<-rep(Price_Fish_Oz,n)
V_Price_Miso<-rep(Price_Miso,n)
V_Entry_Fee<-rep(Entry_Fee,n)
V_Fish_Entitled_Oz<-rep(Fish_Entitled_Oz,n)

set.seed(12)
V_Attendance<-round(rtriang(n,20,50,30),0)
Consumption<-V_Attendance*V_Fish_Entitled_Oz
Available<-V_Order_Oz

V_Profit<-pmin(Consumption,Available)/V_Fish_Entitled_Oz*V_Entry_Fee-V_Order_Oz*V_Price_Fish

#Average profit if they order 160 oz of fish.
mean(V_Profit)
```

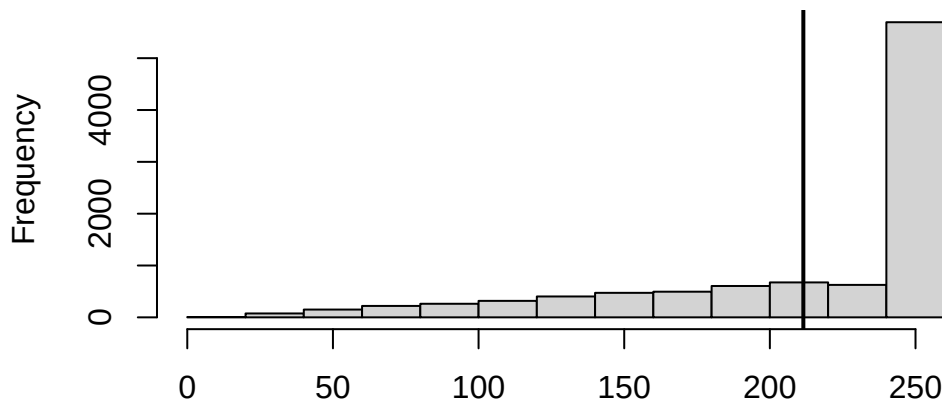
```
[1] 211.498
```

Flaw of Averages

Is there an amount of fish that would give them a higher expected profit?

```
# Generate histogram of simulation results. *Recommendation should be closer to $250.
hist(V_Profit, main="Expected Profits of an Order of 160 Ounces",
      xlab="")
abline(v=mean(V_Profit), lwd=2)
```

Expected Profits of an Order of 160 Ounces



Optimal Order Amount

```
#Optimal order amount. Substitute possible order options and use a look to find highest expected profit
Order_Oz=seq(160,240,10)
Price_Fish_Oz<-9/16 # Input
Price_Miso<-300 # Input
Entry_Fee<-20 # Input
Fish_Entitled_Oz<-5 #Input

Profits<-c()

for (i in Order_Oz){
  n<-100000
  V_Order_Oz<-rep(i,n)
  V_Price_Fish_Oz<-rep(Price_Fish_Oz,n)
  V_Price_Miso<-rep(Price_Miso,n)
  V_Entry_Fee<-rep(Entry_Fee,n)
  V_Fish_Entitled_Oz<-rep(Fish_Entitled_Oz,n)

  set.seed(12)
  V_Attendance<-round(rtriang(n,20,50,30),0) # Random Variable
```

```

Consumption<-V_Attendance*V_Fish_Entitled_Oz # Calculated
Available<-V_Order_Oz # Calculated

V_Profit<-pmin(Consumption,Available)/V_Fish_Entitled_Oz*V_Entry_Fee-V_Order_Oz*V_Price_Fish
Profits<-c(Profits,mean(V_Profit))
}

(results<-data.frame(Order=Order_Oz,Profits=Profits))

```

	Order	Profits
1	160	212.0358
2	170	225.7342
3	180	235.1670
4	190	240.7776
5	200	243.1558
6	210	242.8690
7	220	240.4326
8	230	236.4392
9	240	231.4190

Sensitivity Analysis

Exercises:

Question 1:

Apple is preparing to launch a special edition smartphone accessory designed exclusively for its latest iPhone. This accessory is only compatible with this specific model, and the company anticipates that demand will peak during the first 12 months following the smartphone's release. After this period, a new model of the smartphone is expected to be released, rendering the accessory obsolete. The company estimates that demand for the accessory during the twelve-month period is governed by the probability distribution in the table below.

The production involves a fixed cost of 20,000 dollars, with a variable cost of 15 dollars per unit produced. Each accessory is sold for 100 dollars. However, any unsold units after the twelve-month period will have no value, as they are not compatible with the newest iPhone.

The company is considering producing 10,000 units of the accessory. Using a simulation with 100,000 replications, calculate the expected profit and standard deviation. Additionally, determine the range within which the company can be 90% certain that the actual profit from selling the accessory during the twelve-month period will fall.

```

#Inputs
fixed_cost <- 20000
unit_cost <- 15
unit_price <- 100
units <- 10000

#Random Variable
dem_probs <- c(.1, .2, .3, .25, .15)
dem_val <- c(6000, 7000, 8000, 9000, 10000)

#Simulation
n<-100000
v_fixed_cost<-rep(fixed_cost,n)
v_unit_cost<-rep(unit_cost,n)
v_unit_price<-rep(unit_price,n)
v_units<-rep(units,n)

v_demand<-sample(dem_val, n, replace = TRUE, prob = dem_probs)
revenue<-v_demand*v_unit_price
costs<-v_units*v_unit_cost + v_fixed_cost

v_profit<-revenue-costs
mean(v_profit)

```

```
[1] 645134
```

```
sd(v_profit)
```

```
[1] 119834.5
```

```
quantile(v_profit, c(.05, .95))
```

```

      5%      95%
430000 830000

```

Question 2:

Alaska Airlines is preparing for the peak holiday travel season and needs to decide how many seats to overbook on the SEA-SFO route. Overbooking is expected in the airline industry, as

some passengers typically do not show up for their flights. The airline has data indicating that the number of no-show passengers on this route follows a normal distribution with a mean of 20 and a standard deviation of 5.

If the airline does not overbook enough seats, empty seats result in lost revenue, as the cost of an unused seat is estimated to be \$300. However, if the airline overbooks too many seats, it may need to compensate bumped passengers with vouchers worth \$600 each. Use 10,000 simulations to determine the optimal number of seats to overbook to minimize the airline's expected cost. Try possible values from 10 to 30 in increments of 5.

```
#Inputs
seats <- seq(10, 30, 5)
empty <- 300
overbook <- 600
n <- 10000

Costs<-c()

for (i in seats){
  v_seats<-rep(i,n)
  v_empty<-rep(empty,n)
  v_overbook<-rep(overbook,n)

  no_shows<-rnorm(n, 20, 5)

  empty_seats <- pmax(0, v_seats - no_shows)
  overbook_seats <- pmax(0, no_shows - v_seats)

  empty_cost<-empty_seats*v_empty
  overbook_cost<-overbook_seats*v_overbook

  total_cost<-empty_cost + overbook_cost

  Costs<-c(Costs,mean(total_cost))
}

(results<-data.frame(seats,Costs))
```

	seats	Costs
1	10	6033.786
2	15	3408.263
3	20	1819.650

4	25	1888.896
5	30	3053.741

Question 3:

The Bronco Wine Company is planning to launch a special edition of its finest wine for the upcoming holiday season. The winery faces uncertainty in both customer demand and grape yield, which affects production.

Demand is modeled by a triangular distribution with a minimum of 500, most likely 1,500, and a maximum of 2,500 bottles. Production Costs are 10,000 dollars fixed plus 15 dollars per bottle. Selling Price is 100 dollars per bottle. Salvage Value is 30 per unsold bottle (up to 1,000 bottles). The maximum number of bottles that can be produced is also uncertain and follows a Beta distribution with parameters $\alpha = 2$ and $\beta = 5$, scaled to a maximum capacity of 3,000 bottles. The winery is considering production levels ranging from 1,000 to 3,000 bottles in increments of 500.

Use 10,000 simulations to determine the optimal number of bottles to produce to maximize the winery's expected profit. Create a five number summary to describe the distribution of profits at the optimal number of bottles.

```
#Demand = triangular distribution
#Capacity = beta distribution
#Order_quantity = MIN(capacity, order)
#Costs = production_cost + unit_cost*order_quantity
#Revenue = unit_price*MIN(demand, order_quantity) + sal_price*MIN(MAX(0,order_quantity - demand), 1000)
#Profits = Revenue - Costs

order <- seq(1000, 3000, 500)
production_cost <- 10000
unit_cost <- 15
unit_price <- 100
sal_price <- 30

Profits <- c()
Min_Profits <- c()
Q1_Profits <- c()
Q2_Profits <- c()
Q3_Profits <- c()
Max_Profits <-c()

for (i in order) {
```

```

n<-10000
v_order<-rep(i,n)
v_production_cost<-rep(production_cost,n)
v_unit_cost<-rep(unit_cost,n)
v_unit_price<-rep(unit_price,n)
v_sal_price<-rep(sal_price,n)

demand<-rtriang(n, 500, 2500, 1500)
capacity<-rbeta(n, 2, 5)*3000

order_quantity<-pmin(i, capacity)
costs<-v_production_cost + (order_quantity*v_unit_cost)
revenue<-v_unit_price*pmin(demand, order_quantity) + v_sal_price*pmin(pmax(0,order_quantity), order_quantity)
profit<-revenue - costs

Profits<-c(Profits, mean(profit))
Min_Profits <- c(Min_Profits, min(profit))
Q1_Profits <- c(Q1_Profits, quantile(profit, probs=.25))
Q2_Profits <- c(Q2_Profits, quantile(profit, probs=.5))
Q3_Profits <- c(Q3_Profits, quantile(profit, probs=.75))
Max_Profits <-c(Max_Profits, max(profit))
}

(results<-data.frame(order,Profits,Min_Profits,Q1_Profits,Q2_Profits,Q3_Profits,Max_Profits))

```

	order	Profits	Min_Profits	Q1_Profits	Q2_Profits	Q3_Profits	Max_Profits
1	1000	50778.98	-9919.169	31929.95	56127.34	75000.00	75000.0
2	1500	57780.14	-8551.013	30928.68	56379.74	83191.03	117500.0
3	2000	58218.21	-8703.102	30393.57	55732.04	82937.45	160000.0
4	2500	58575.51	-8929.651	31212.14	56156.39	83145.90	187613.6
5	3000	57974.59	-9505.361	30892.62	55185.73	81820.97	176007.3

```

#Decision variable
levels<-seq(1000, 3000, 500)

#Inputs
fixed_cost<-10000
unit_cost<-15
unit_price<-100
salv_price<-30 #Up to 1000 units

Profits<-c()

```

```

for (i in levels) {
  n<-10000
  v_levels<-rep(i,n)
  v_fixed_cost<-rep(fixed_cost,n)
  v_unit_cost<-rep(unit_cost,n)
  v_unit_price<-rep(unit_price,n)
  v_salv_price<-rep(salv_price,n)

  #Random variables
  demand<-rtriang(n, 500, 2500, 1500)
  capacity<-rbeta(n, 2, 5)*3000

  #Calculated values
  production<-pmin(i, capacity)
  sales<-pmin(production, demand)
  leftover<-pmax(0, production - sales)

  revenue<-sales*v_unit_price
  cost<-v_fixed_cost + production * v_unit_cost
  salv_revenue<- pmin(leftover, 1000) * v_salv_price
  profit<-revenue + salv_revenue - cost

  Profits<- c(Profits, mean(profit))
}

(results<-data.frame(levels,Profits))

```

	levels	Profits
1	1000	50657.13
2	1500	57222.34
3	2000	58948.24
4	2500	59183.51
5	3000	59181.20

Question 4:

Question 5: